

Fast Ethernet:

- Fast Ethernet was designed to compete with LAN protocols such as FDDI (Fiber Distributed Data Interface) or Fiber Channel.
- IEEE created Fast Ethernet under the name 802.3u.
- Fast Ethernet is backward-compatible with Standard Ethernet.
- It can transmit data 10 times faster at a rate of 100 Mbps.

MAC Sub layer

- A main consideration in the evolution of Ethernet from 10 to 100 Mbps was to keep the MAC sub layer untouched. However, a decision was made to drop the bus topologies and keep only the star topology. For the star topology, there are two choices: half duplex and full duplex. In the half-duplex approach, the stations are connected via a hub; in the full-duplex approach, the connection is made via a switch with buffers at each port.
- The access method is the same (*CSMA/CD*) for the half-duplex approach; for full duplex Fast Ethernet, there is no need for *CSMA/CD*.
- However, the implementations keep *CSMA/CD* for backward compatibility with Standard Ethernet.

Auto negotiation:

A new feature added to Fast Ethernet is called auto negotiation. It allows a station or a hub a range of capabilities. Auto negotiation allows two devices to negotiate the mode or data rate of operation.

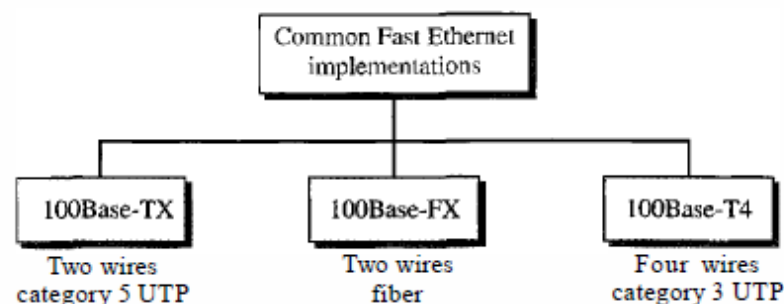
It was designed particularly for the following purposes:

- To allow incompatible devices to connect to one another. For example, a device with a maximum capacity of 10 Mbps can communicate with a device with a 100 Mbps capacity (but can work at a lower rate).
- To allow one device to have multiple capabilities.
- To allow a station to check a hub's capabilities.

Physical layer:

- Two topology: star or point to point.

Implementation:



Gigabit Ethernet:

- The need for higher data rate resulted in the design of the Gigabit Ethernet (1000 Mbps).
- The IEEE committee calls the standard 802.3z.

- Upgraded to the data rate to 1 Gbps.
- Compatible with Standard or Fast Ethernet.

MAC layer:

- Uses the same 48-bit address.
- Uses the same frame format.
- It permits to keep the same minimum and maximum frame lengths.
- It supports auto negotiation as defined in Fast Ethernet.
- For the star topology, there are two choices: half duplex and full duplex
- The access method is the same (*CSMA/CD*) for the half-duplex approach; mostly full duplex mode is used.
- If half duplex mode is used then three possible approaches:

Traditional:

In the traditional approach, we keep the minimum length of the frame as in traditional Ethernet (512 bits). However, because the length of a bit is 100 times shorter in Gigabit Ethernet than in 10-Mbps Ethernet, the slot time for Gigabit Ethernet the maximum length of the network is 25 m for the same minimum frame size. This length may be suitable, if all the stations are in one room, but it may not even be long enough to connect the computers in one single office.

Carrier Extension:

To allow for a longer network, we increase the minimum frame length. The carrier extension approach defines the minimum length of a frame as 512 bytes (4096 bits). This means that the minimum length is 8 times longer. This method forces a station to add extension bits (padding) to any frame that is less than 4096 bits. In this way, the maximum length of the network can be increased 8 times to a length of 200 m. This allows length of 100 m from the hub to the station. Carrier extension, essentially tells the hardware to add its own padding after the normal frame to extend the frame to 512 bytes. Since this padding is added by the sending hardware and removed by the receiving hardware, the software is unaware of it, meaning that no changes are needed to existing software.

Frame Bursting:

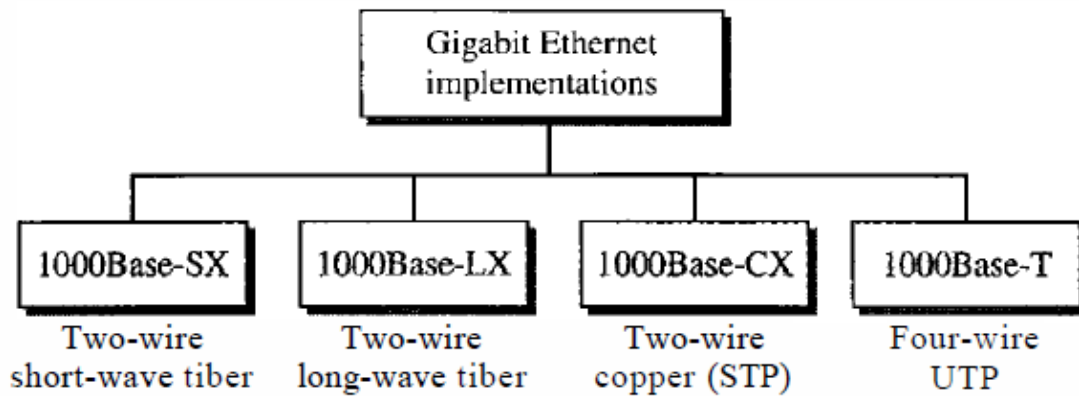
Carrier extension is very inefficient if we have a series of short frames to send; each frame carries redundant data. To improve efficiency, frame bursting was proposed. Instead of adding an extension to each frame, multiple frames are sent. In other words, the method deceives other stations into thinking that a very large frame has been transmitted.

- CSMA/CD not used in full duplex mode.

Physical layer

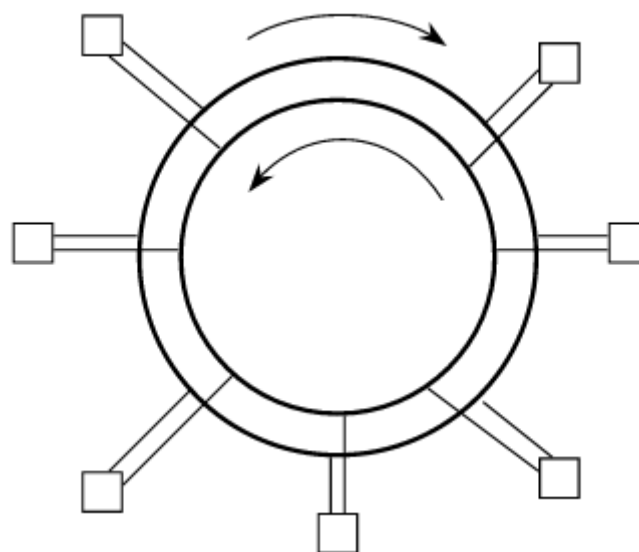
Implementation

Gigabit Ethernet can be categorized as either a two-wire or a four-wire implementation. The two-wire implementations use fiber-optic cable (1000Base-SX, short-wave, or 1000 Base-LX, long-wave), or STP (1000Base-CX). The four-wire version uses category 5 twisted-pair cable (1000Base-T). In other words, we have four implementations, 1000 Base-T was designed in response to those users who had already installed this wiring for other purposes such as Fast Ethernet or telephone services.



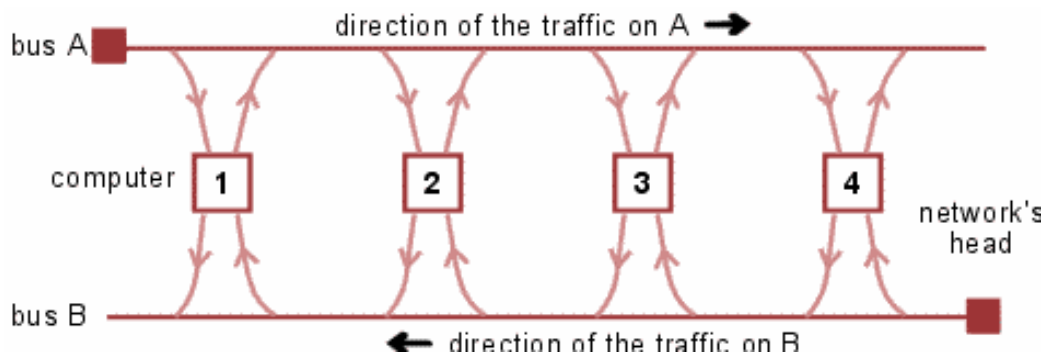
FDDI:

- FDDI was developed by the American National Standards Institute (ANSI) X3T9.5 standards committee in the mid-1980s
- It uses a rotating token to define which system can send data at any given time.
- In FDDI, token is absorbed by station and released as soon as it completes the frame transmission
- The Fiber Distributed Data Interface (FDDI) specifies a 100-Mbps token-passing, dual-ring LAN using fiber-optic cable.
- An FDDI network contains two token rings, one for possible backup in case the primary ring fails.
- A FDDI dual ring supports up to 500 nodes per ring. The total maximum length for each of the cable rings is 100 km with nodes up to 2 km apart(max limit) on multimode fiber, and 10 km apart on single-mode fiber.
- FDDI frame format is similar to token ring 802.5
- If the Primary ring fails and the FDDI wraps, then the network design must be such that the total ring length must not exceed 200km and the number of nodes must not exceed 1000.
- In addition, variation of FDDI, called FDDI full duplex technology (FFDT) uses the same network infrastructure but can potentially double data rates. If the secondary ring is not needed for backup, it can also carry data, extending the network's capacity to 200 Mb/s.

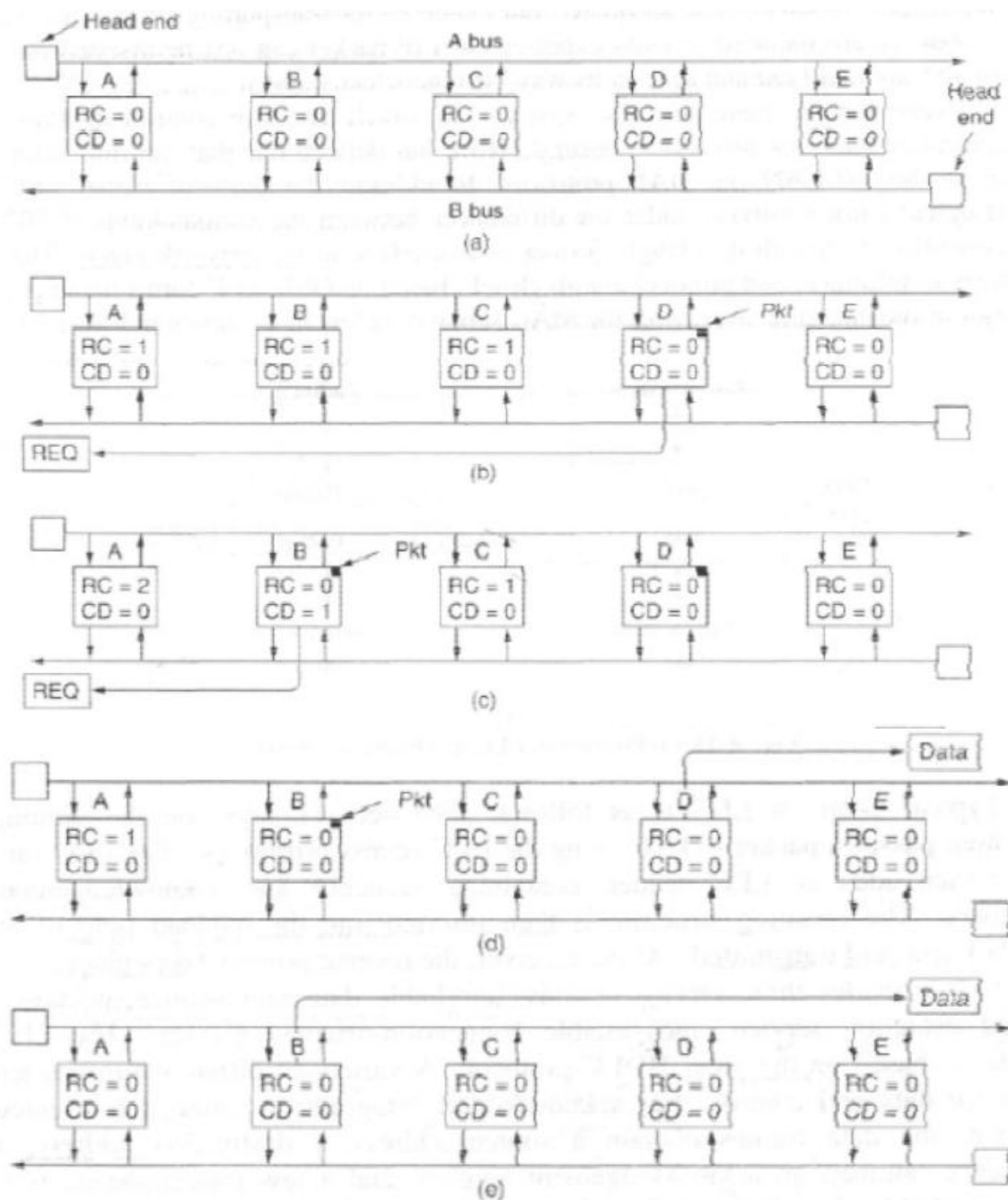


IEEE 802.6:

- IEEE defined a MAN standard for covering an entire city. This was called Distributed Queue Data Interface (DQDB) and put up as IEEE standard 802.6.
- The basic geometry of DQDB is as shown in the figure below. Two parallel unidirectional buses are laid down in the area to be covered by the network. The stations are attached to both the buses in parallel.



- Each bus has a head-end, which generates a steady stream of 53 byte cells.
- Each cell travels downstream from the head end. When it reaches the end, it falls off the bus. Each cell carries a 44 byte payload field.
- Each cell also holds two protocol bits, Busy, set to indicate that a cell is occupied, and Request, which can be set when a station wants to make a request.
- To transmit a cell, a station has to know whether the destination is to the left of it or to its right. If the destination is to the right, the sender uses bus A. Otherwise it uses bus B.
- The 802.6 protocol does not have greedy stations. In all others, if a station gets a chance to send, it sends.
- Here the queue up and become ready to transmit in FIFO order, all this without having a central queue. To simulate the FIFO queue, each station maintains two counters, RC and CD. RC (Request Counter) counts the number of downstream request pending until the station itself has a frame to send. At that point, RC is copied to CD, RC is reset to 0 and now counts the number of requests made after the station became ready.
- For example, if $CD = 3$ and $RC = 2$ for station k, the next three empty cells that pass by station k are reserved for downstream stations, then station k may send, then two more cells are reserved for downstream stations. For simplicity, we assume a station can have only one cell ready for transmission at a time.



- To send a cell, a station must first make a reservation by setting the Request bit in some cell on the reverse bus (i.e., on bus B for a transmission that will later take place on bus A).
- As this cell propagates down the reverse bus, every station along the way notes it and increments its RC.
- To illustrate this concept, we will use an example. Initially, all the RC counters are 0, and no cells are queued up, as shown in Fig.(a).
- Then station D makes a request, which causes stations, C, B, and A, to increment their RC counters, as shown in Fig.(b). After that, B makes a request, copying its current RC value into CD, leading to the situation of (c).
- At this point, the head-end on bus A generates an empty cell. As it passes by B, that station sees that its $CD > 0$, so it may not use the empty cell. (When a station has a cell queued, CD represents its position in the queue, with 0 being the front of the queue).
- Instead it decrements CD. When the still-empty cell gets to D, that station sees that $CD = 0$, meaning that no one is ahead of it on the queue, so it puts its data into the cell and sets the Busy bit.

- After the transmissions are done, we have the situation of Fig.(d).
- When the next empty cell is generated, station B sees that it is now at the head of the queue, and seizes the cell as illustrated in fig.(e).
- In this way, stations queue up to take turns, without a centralized queue manager hence queue is known as distributed queue.

Technical Facts Of DQDB

- Distance up to 200 KM
- Medium: Copper or Fiber
- At distance up to 160 KM approx speed is 44.73 Mbps (Copper)
- At distance up to 100 KM approx speed is 150 Mbps (Fiber)

DQDB Features

- DQDB is a MAC layer protocol for MAN
- Unlike FDDI, DQDB is an IEEE standard: 802.6
- Designed for both voice & video
- Topology used: Dual Bus - uses 2 unidirectional logical buses